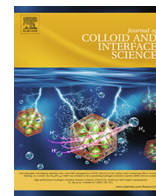




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Regular Article

Insights from molecular dynamics simulations on structural organization and diffusive dynamics of an ionic liquid at solid and vacuum interfaces



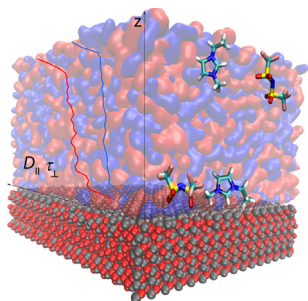
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GRAPHICAL ABSTRACT



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ABSTRACT

Hypothesis: A reliable modelling approach is required for simultaneous characterisation of static and dynamic properties of bulk and interfacial ionic liquids (ILs). This is a prerequisite for a successful investigation of experimentally inaccessible, yet important properties, including those that change significantly with the distance from both vacuum and solid interfaces.

Simulations: We perform molecular dynamics simulations of bulk [C₂Mim][NTf₂], and thick IL films in contact with vacuum and hydroxylated sapphire surface, using the charge methods CHelpG, RESP-HF and RESP-B3LYP with charge scaling factors 1.0, 0.9 and 0.85.

Findings: By determining and employing appropriate system sizes and simulations lengths, and by benchmarking against self-diffusion coefficients, surface tension, X-ray reflectivity, and structural data, we identify RESP-HF/0.9 as the best non-polarizable force field for this IL. We use this optimal parametrisation to predict novel physical properties of confined IL films. First we fully characterise the internal

Abbreviations: ILs, Ionic liquids; MD, molecular dynamics; ff, force field; S-L-V, solid-liquid-vacuum; V-L-V, vacuum-liquid-vacuum; S-L, solid-liquid; L-V, liquid-vacuum; S-L-S, solid-liquid-solid; SILP, Supported Ionic Liquid Phase catalysts; SCILL, Solid Catalysts with Ionic Liquid Layer; CL&P, Canongia Lopes and Pádua force field; CHelpG, Charges from Electrostatic Potentials using a Grid; RESP, Restrained Electrostatic Potential; INND, interface-normal number density; SI, Supplementary Information; rdf, radial distribution functions; COM, centre of mass; IFF, INTERFACE force field; DiB, Data in Brief Manuscript.

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X-Ray reflectivity
Structural order
Lateral diffusion
Residence time

configurations and orientations of IL molecules relative to, and as a function of the distance from the solid and vacuum interfaces. Second, we evaluate densities together with mobilities in-plane and normal to the interfaces and find that strong correlations between the IL's stratification and diffusive transport in the interfacial layers persist for several nanometres deep into IL films.

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1. Introduction

Ionic liquids (ILs) are molten organic salts with complex cations and anions that make them liquids even below 100 °C [1]. ILs are well known as designer solvents due to their unique properties, such as non-volatility, tuneable polarity, high viscosity, electrochemical stability and high solvent capacity [2,3]. The large interest within the scientific community, as well as in an industrial setting, is demonstrated by inspection of their diverse applications, as evident from a steady increase in the number of scientific publications and patents [4]. The most common applications of ILs are as electrolytes [5,6] in batteries [4,7–12], supercapacitors [13–15], lubricants [16–21], in tissue preservation [22], separations and purification of liquids from metal [23,24], biomass conversion [25] and smart films [26,27]. Fairly recently ILs are in the spotlight in the field of surface engineering [28], catalysis and synthesis [29–33].

It is in this latter connection that new approaches to enhance the activity of the catalysts combined with the possibility of reusing IL surface catalysis, have been developed. Pertinent examples are Supported Ionic Liquid Phase (SILP) catalysts [34–36] and Solid Catalysts with Ionic Liquid Layer [37,38] (SCILL). In both of these approaches, the IL of choice is a prototypical liquid. Applications of these approaches can be found in hydroformylation, hydrogenation, carbonylation and fine chemical synthesis [39]. Both aforementioned techniques are based on depositing a thin film of ionic liquid on heterogeneous support materials belonging to a porous particle. While in the SILP technology, homogeneous transition metal complexes are dissolved within the IL film, in the case of SCILL, the IL surrounds a heterogeneous solid catalyst. From these studies it has become clear that the choice of the exact type of the IL employed is a crucial parameter for a successful realization of the catalytic process. In particular, the choice of both cation and anion can have major effects on vital properties such as the solubility of reactants and gases transported through the pore and across the liquid-vacuum (L-V) interface toward the catalyst. Furthermore, structuring of the IL at the solid-liquid (S-L) interface [40] and the influence of the IL on the catalyst may play a crucial role on the overall reaction turnover, which further motivates the need to optimize the choice of IL, with particular respect to transport and organisation of the film.

Molecular dynamics (MD) simulations are a powerful tool capable of providing much needed insight into structural ordering of the IL at both S-L and L-V interfaces [41–44]. However, even for studying a chosen IL in the bulk, the first challenge in modelling is to find the correct force field (*ff*), which describes both static and dynamic properties of the IL well. In a recent review [45], an overview of all significant efforts made in the last 15 years in this direction is given. Models that include explicit polarization, which are parameterized using a number of experimental results (including densities and diffusion coefficients) [46,47] or from *ab initio* calculations [48], appear to be associated with the highest level of accuracy. This has been found to be particularly important when computing, for example, solute-solvent interaction energies and related properties [49]. However, for simulations with more than 1000 ion pairs, such an approach is still computationally very

demanding [45]. For this reason, non-polarizable (fixed-charge) force fields continue to enjoy widespread application. Indeed, such models appear capable of producing reliable results for the overall investigation of structural and dynamic properties. In particular, the introduction of charge-scaling methods [50], as an approximate, mean-field manner in which to account for polarization effects, can produce results of a quality comparable to explicitly polarizable-ion models [51].

Hence, currently the choice of an all-atom non polarizable classical *ff* is found most suitable when simulating an IL with a large number of ion pairs. Here, the most widely used force field is the one developed by Canongia Lopes and Pádua (CL&P) [52–55], mostly because a large combination of typical cations and anions found in ILs have been parametrized. This *ff* applies bonded and van der Waals terms from OPLS-AA or AMBER *ff*, thereby employing the *CHarges from electrostatic potentials using a Grid* (CHelpG) method to obtain atomic charges. Among various methods that exist in the literature for assigning atomic charges within standard force fields [56,57], CL&P *ff* is one method that gives uniformly satisfactory results for calculated densities and structural properties [52]. In order to improve the description of dynamic and transport properties, Maginn [58–63] and collaborators used the *Restrained Electrostatic Potential* (RESP) method to obtain partial charges and developed a *ff* based on CL&P, which was used for uniform charge scaling to mimic polarization effects in their simulations. For completeness it is worth mentioning here that if the focus of the simulations is mainly on dynamic properties, an alternative method to downscaling of the charges has been proposed by Kodderman [64,65], where the Lennard-Jones parameters are changed or scaled instead.

When setting up our simulations with the aim to study the IL in an environment representative of SILP and SCILL prior to introducing a catalyst, implying over 10 nm thick layers of IL, a further and more serious challenge is to find an optimal force field that yields realistic structuring and interfacial orientation of the ions at the S-L and L-V interfaces, which are in reasonable agreement with experimental results. Simulations in this direction started only in the last decade [66–68], and require a correct parametrisation of the interaction of IL with the interface. To this end, one attractive approach was developed by Heinz and collaborators and cast in the framework of the *INTERFACE force field* (IFF) for inorganic-organic interfaces. Their approach uses an integrated way of assigning both charges and Lennard-Jones parameters of the solid and the IL to validate various thermodynamic and surface properties [69]. Because of this, IFF is particularly useful for simulating isolated molecules or very thin films [41], but its application is not evident in the context of thick IL layers. For our case, also relevant to SILP and SCILL, several alternative force-fields were developed [70,71]. In particular, CLAYFF was shown to agree with DFT in calculations of structural, vibrational and thermodynamic properties of metal-oxides [72]. This success established CLAYFF as a common choice modelling insulators [73], although its combination with the IL force fields is still subject to debate.

In a previous work from our group, using an archetypical model system consisting of an imidazolium-based IL ([C₂Mim][NTf₂]) in contact with a fully hydroxylated sapphire substrate, we presented

and compared the results of high-resolution X-ray reflectivity measurements and extensive MD simulations with IL in a solid-liquid-solid (S-L-S) configuration [74]. We found using the experimental X-ray reflectivity as a clear benchmark that the Maginn *ff* (with CHelpG charges) for the cation and the CL&P *ff* for the anion was more preferable than the approach proposed by Kodderman and that the Lorentz-Berthelot mixing rule produces results superior to those from the geometric mixing rule, when coupling sapphire with the chosen IL.

We build on this previous effort to now investigate which parametrization can simultaneously recover properties of the bulk liquid, as well as of the liquid at the interface with the solid and the vacuum. We investigate in detail the effect of different charge methods (CHelpG [75], RESP-HF [76] and RESP-B3LYP [77]) and that of charge scaling (0.85–1.0) in three model systems shown in Scheme 1 (pure liquid (L), vacuum-liquid-vacuum (V-L-V), solid-liquid-vacuum (S-L-V)). First we establish the minimal requirements for proper consistency (system geometry) and convergence (simulation time) essential for describing structural and dynamic properties of ILs as a function of confinement, which has not been fully appreciated in the literature to date. Second, we determine a common parametrization that most accurately reproduces available measurements in all three geometries, and finally, we apply it to study spatially-dependent properties of stratified liquids, which are at this stage accessible only to molecular modelling but important for understanding experimental observations.

2. Simulation details

2.1. Parametrization

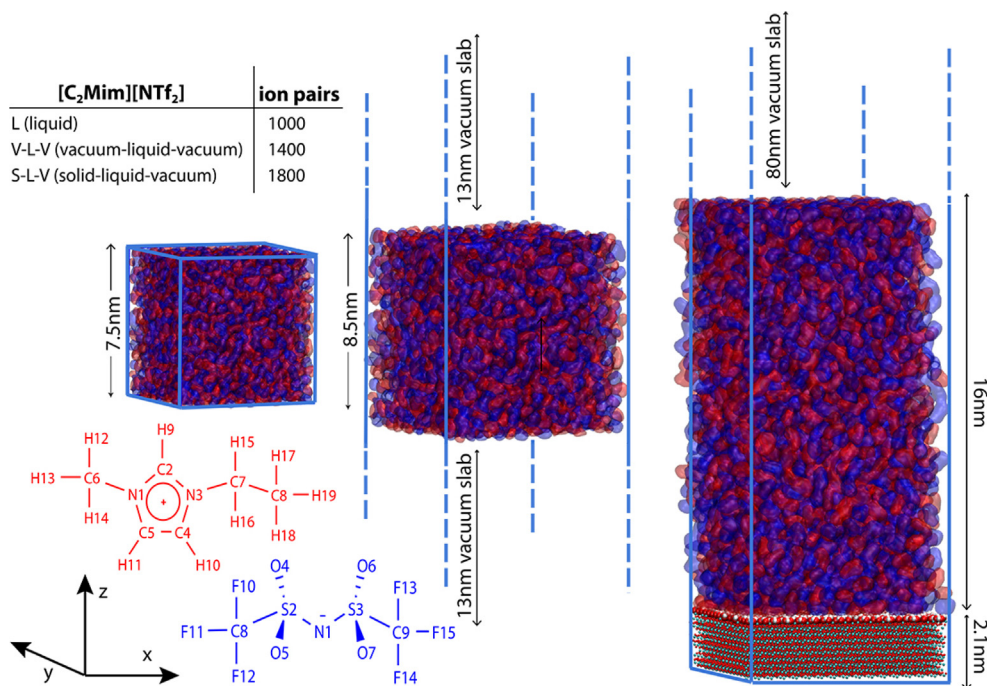
We use three different parametrization schemes for the force field: (a) Maginn [61]/CL&P [52] parameters with CHelpG charges [75], (b) Maginn/CL&P parameters with RESP-HF charges (HF/6-31G(d) level of theory) [76] and (c) Maginn/CL&P parameters with

RESP-B3LYP charges (B3LYP/cc-pVTZ level of theory) [77]. In all three parametrization schemes, the Maginn parameters were used for the cation, the CL&P were used for the anion, and the Lorentz-Berthelot mixing rule [78,79] was applied. The original atomic charges were rescaled to 90% and 85% of their initial values, leading to a total of nine different sets of parameters. We performed MD simulations using each of these sets for the three previously mentioned systems (L, V-L-V, and S-L-V). Below we give details of the preparation of the three systems. All simulations were performed in GROMACS 5.1.2 [80] with a time step of 2 fs and a cut-off of 2 nm for the van der Waals and short-range Coulomb interactions. Three-dimensional periodic boundary conditions were employed, along with the particle-mesh Ewald procedure for a proper description of the long-range Coulomb interactions for all three model systems.

2.2. Systems

Bulk liquid (L): The prepared “pure” IL system contains 1000 pairs of cations $[\text{C}_2\text{Mim}]^+$ and anions $[\text{NTf}_2]^-$ (see Scheme 1). The starting conformations for anion were both *cis* and *trans*. The systems were first minimized, relaxed via the NVT ensemble for 5 ns and then equilibrated using the NPT ensemble for an annealing procedure for 12 ns ($P = 1 \text{ atm}$, $\beta = 4.8 \times 10^{-5} \text{ MPa}^{-1}$) using the Berendsen barostat [81]. The annealing protocol was as follows: 3 ns from 300 K to 700 K, 3 ns at 700 K, 4 ns from 700 K to 300 K and finally 2 ns at 300 K with the Nosé-Hoover thermostat [82,83]. The obtained model system was subsequently simulated under NVT conditions (cubic box of side length 7.5 nm, density 1520 kg/m^3) at $T = 300 \text{ K}$ for 100 ns, with the temperature being controlled using the Nosé-Hoover thermostat.

Liquid with vacuum interfaces (V-L-V): Here, at first the pure ionic liquid was annealed using the same procedure described for the L system. To this fully equilibrated IL (1400 ion pairs in a cubic box of side length 8.5 nm, density 1520 kg/m^3), a vacuum slab (13 nm) was added on each side, thus defining the V-L-V system



Scheme 1. Model systems: pure liquid (L), vacuum-liquid-vacuum (V-L-V), and solid-liquid-vacuum (S-L-V). The chosen cation $[\text{C}_2\text{Mim}]^+$ (red) and anion $[\text{NTf}_2]^-$ (blue) are represented schematically. Analogous figure but in ball-and-stick configuration is shown in Data in Brief [89] Fig. D1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(see Scheme 1). In both cases, the entire system was first minimized and followed by NVT simulations at $T = 300$ K for 200 ns. Since some time is required to establish the interface, only the last 100 ns were used for analysis. The Nosé-Hoover thermostat was used to keep the temperature constant. In addition, a simulation with 1000 ion pairs was performed using the analogous protocol to match the L system, but in this case no bulk is found in the V-L-V geometry.

Liquid film (S-L-V): The simulated S-L-V system consisted of a slab of sapphire ($7.57 \text{ nm} \times 6.29 \text{ nm} \times 2.12 \text{ nm}$) optimized in GULP [84] with a fully hydroxylated (0001) x-y surface, described by the CLAYFF [85] force field. We placed 1800 ion pairs of IL (equilibration in a monoclinic box $7.57 \text{ nm} \times 6.29 \text{ nm} \times 16 \text{ nm}$, density 1520 kg/m^3) above the surface of the sapphire (2.1 nm) in a monoclinic simulation box as in our previous work [74]. Coupling between the IL and the sapphire *ff* was obtained using Lorentz-Berthelot mixing rules. The systems were first minimized and then semi-isotropic NPT annealing simulations, using $P = 1 \text{ atm}$, $\beta = 4.8 \times 10^{-5} \text{ MPa}^{-1}$, were performed for 12 ns, the annealing protocol being the same as for the L system. Further, the box size was allowed to vary in the z direction. Then, an 80 nm large vacuum was placed above the IL-sapphire system. The resulting model system is shown in Scheme 1. Thus, the model system spans a height of about 100 nm in the z-direction. This system was simulated using the NVT ensemble at $T = 300$ K, with the temperature being controlled using the Nosé-Hoover thermostat using the Langevin algorithm. From a 360 ns run, the first 100 ns were required for complete equilibration, 160 ns and 260 ns were used for converging averaged and spatially dependent properties, respectively.

Note that the height of the vacuum in the S-L-V system is more than ten times larger than the dimensions along the x- or y-directions. The presence of this large vacuum results in minimizing the contribution of the z-replicas to the electrostatic interactions in the central simulation cell [86]. Thus, if we were to estimate either the electrostatic potential along the z-direction or the Coulomb forces between any two ions in the z-direction, we could expect to obtain an estimation comparable to that resulting from an Ewald sum in the slab geometry [87,88]. We can therefore safely conclude that the interfacial ordering of the ions of the IL at the substrate and their diffusion along z-direction is dictated primarily by the electrostatic interactions within the central simulation cell. We believe that using the large vacuum as a means to reduce the above mentioned effects conveys a further distinct advantage for

examining the true effects of charge scaling on the structural ordering and dynamics of the IL in the S-L-V model system.

3. Results

3.1. Effect of confinement of the relaxation and convergence of bulk liquid properties

As we wish to compare and contrast systems comprised of solid, liquid and vacuum regions, as well as the interfaces between them (Scheme 1), it is important to consider each system on an equal footing. In particular, in order to meaningfully differentiate the properties of the interfaces from the properties of the bulk liquid in each system, it is paramount to be able to recover the properties of the bulk liquid in each case. Only under the restrictions of such a criterion, one can be reasonably confident that any structural organization of the IL at the S-L and L-V interfaces is genuine and that the interfaces themselves are well separated and free of influence from one another. In this light, we present below a brief analysis of the conditions required to reproduce the fully mixed liquid state of the bulk IL in the presence of the S-L and L-V interfaces for the RESP-HF/0.9 force field (see Fig. 1).

System size: While bulk liquid properties are often gauged through pairwise correlation functions (such as radial distribution functions, see Fig. D4 in Data in Brief (DiB) [89]), an additional convenient measure in the present context is the interface-normal number density (INND, Fig. 1, right panel). In the L system, with any arbitrarily defined interface plane, we find that 1000 ion pairs is sufficient to eliminate spatial fluctuations in the INND (Fig. 1, top right) and produce a fully mixed bulk state.

The introduction of two vacuum interfaces to the L system, as in the V-L-V system, has a relatively strong effect on the INND (see Fig. D2 [89]). Indeed, with 1000 ion pairs, the homogeneity of the region with minimal INND fluctuations, of some 2.5 nm in width, is not entirely clear. The system with 1400 increases the extent of mixing of the homogenised bulk region, which corresponds to roughly 4 nm (see Fig. 2a and b [89]) or approximately half the thickness of the IL layer in the V-L-V system (cf. Scheme 1). The V-L-V system with 1400 ion pairs would, therefore, seem to constitute a minimally acceptable size according to the aforementioned criterion and is accordingly used henceforth.

The introduction of a solid-liquid interface, as in the S-L-V system has an even more dramatic effect on the INND (see Fig D2c in

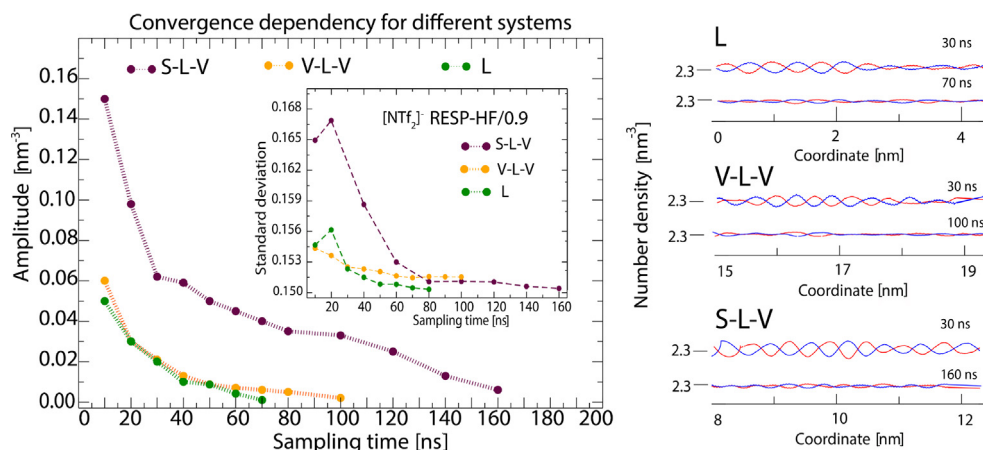


Fig. 1. Left panel: Decrease of interface-normal number density (INND) amplitudes in bulk region for the anion as a function of sampling time. The inset shows the corresponding decrease of the standard deviation. Right panel: INND profile per molecule of cation (red line) and anion (blue line) as a function of coordinate of the bulk region of IL corresponding to Scheme 1. The origin of the coordinate system in INND is chosen by the position of hydroxyl groups on the sapphire surface (S-L-V system), and in the V-L-V and L systems at the edge of the simulation box. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

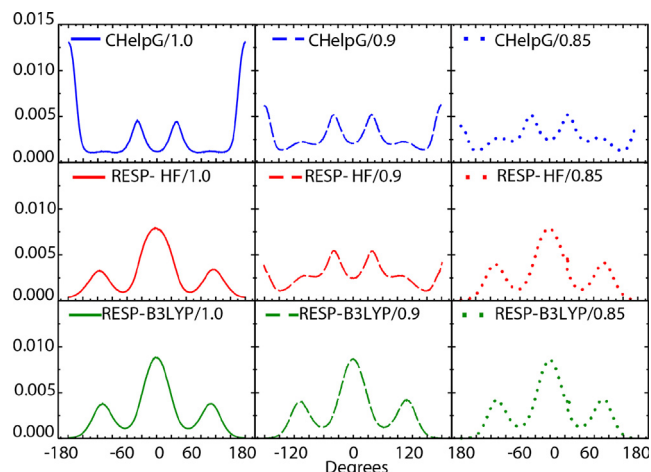


Fig. 2. Distribution of dihedral angle C—S—S—C of $[\text{NTf}_2]^-$ ion. CHelpG method displays a preference for the *trans* conformation for full charge, but the *trans/gauche* conformation with charge scaling (top panel); the RESP-B3LYP method displays a uniformly strong preference for the *cis* conformation irrespective of charge scaling (bottom panel). RESP-HF on the other hand prefer *cis* conformations for full charge and S.F. = 0.85, while a *trans/gauche* conformation is preferred for S.F. = 0.9.

DiB [89] and Ref. [74]). Indeed, it is apparent that a liquid phase of some 7.5 nm (L) or 8.5 nm (V-L-V) would not be sufficiently large to recover a homogenous bulk region and ensure that the interfaces are mutually independent. For this reason, we have chosen to introduce a further 400 ion pairs. The resulting S-L-V system, with 1800 ion pairs and some 16 nm of liquid, exhibits a homogenized bulk region of roughly 7 nm.

Simulation and sampling time: As each of the three systems was being simulated, periodic testing showed that long simulation times were necessary to obtain equilibrated interfaces (see Simulation Details). Further, employing amplitudes of the INND fluctuations, together with their standard deviations as measures of convergence, we found that a fully homogeneous bulk region is achieved only when the sampling (averaging) time corresponds to the last 70 ns, 100 ns and 160 ns for the L, V-L-V and S-L-V systems, respectively (see left panel of Fig. 1). Incidentally, the standard deviation for the three systems also tends to a constant value for the same range of corresponding sampling times.

In the right panel of Fig. 1, we demonstrate that structural ordering in what should be the unstructured bulk region of the IL is clearly visible in the three model systems when only the last 30 ns are used for sampling. In contrast, the bulk region is completely homogenised when the aforementioned sampling times are used. These conclusions are supported by the total in-plane correlation functions calculated in the bulk region for all three systems, shown in Fig. D3 in DiB [89].

Guided by the above findings the rest of the study has been carried out with appropriately large simulation and sampling times. Satisfying the criteria outlined in this section gives us a secure background to perform an objective comparison of the influence of the charge methods and scaling factors on all three model systems.

3.2. Comparative study of force fields

Our interest is in obtaining an accurate description of both bulk and surface properties simultaneously. Since we did not make recourse to any further fitting procedures while calculating the relevant properties, comparison of the results to experiments using all nine parametrizations is expected to reveal the inherent performance of these force fields and the scaling factors in the properties studied.

L - static and dynamic properties: We start by inspecting overall structural organization of the investigated IL by considering the various radial distribution functions (*rdfs*), $g(r)$, of anion-cation ($g_{ac}(r)$), cation-cation ($g_{cc}(r)$) and anion-anion ($g_{aa}(r)$) of the ions which gives insight regarding the average structure of the liquid (Fig. 4). It is well known that unlike other typical liquids, structural correlations in ILs are long-ranged because they are composed of charged species. From literature we know that the shorter the alkyl chains of the IL, the more long-ranged are the correlations [90]. Another important indication of the strong structural ordering due to charge-charge interaction is evident from the structure factors of the molecules, $S(q)$ presented in Fig. S1 in Supplementary Information (SI), for all nine parametrizations, along with experimental structure factors obtained using small and wide angle x-ray scattering (SWAXS) experiments [91].

An inspection of Fig. S1 shows that all nine parametrizations exhibit many following common features. The first peak is centred at approximately 9 nm^{-1} and is associated with cation-cation or anion-anion distances within the polar network. This is characteristic for ILs with an $[\text{NTf}_2]^-$ anion. A second peak is centred at 15 nm^{-1} and is related to the cation and anion distance of direct contact pairs [90].

An inspection of the three panels shows that all three charge methods provide a good description of the structure factor [92]. Any effect of charge scaling is more pronounced in the first peak in the case of RESP-HF/0.9 and is also shifted more to the right. A similar effect of more pronounced peaks is found for RESP-HF/0.9 parametrization, when one inspects the *rdfs* obtained by all three charge methods for a scaling factor of 0.9 of the H9 of the cation (cf. Scheme 1) with nitrogen, sulphurs and oxygens of the anion (Fig. D5 in DiB [89]). While scaling of charges allows for atoms and molecules to approach closer in the liquid environment, it is evident that the strength of these charge interactions depends on the specific charge method. However, we see from Fig. S1 that the behaviour of $S(q)$ for RESP-HF/0.85 is very similar to that of RESP-HF/1.0, indicating that short-range *ff* interactions may offset the stronger charge interactions. As we will see further, this lack of systematic trend seems to be an inherent feature of the IL due to various forces at play in the liquid state.

Another frequently quantified structural property in the gas and liquid phase for the $[\text{NTf}_2]^-$ is C—S—S—C dihedral. In previous studies of ILs containing the $[\text{NTf}_2]^-$ anion, the *trans/gauche* conformation of the dihedral was found to be more common [93], whereas in its crystal phase the *cis* conformation is more dominant [94]. In our system we anticipate both shortness of the alkyl chain of the cation and the close structural ordering of the ions to strongly influence this studied property.

We found, based on our QM calculations (Gaussian09 software [95]), that the preferred conformation of the anion in the gas phase is *trans* (Fig. D6 [89]). Furthermore, we have calculated the dihedral distribution in the liquid phase using MD simulations for all parametrizations. We performed two sets of equilibrations, both NVT and NPT annealing for 12 ns (300 K to 700 K (3 ns), at 700 K (3 ns), 700 K to 300 K (4 ns) and finally at 300 K for 2 ns), also with two different starting configurations of anion conformations (*cis* and *trans*). The distribution of the dihedral was calculated after equilibration, and was continually monitored during production runs, during which it showed no variation. The results are presented in Fig. 2.

We now consider the effect of the charge method and scaling in some detail. From the top left panel of Fig. 2 we find that the CHelpG parametrization with full charges shows the same distribution of the dihedral of the anion as in the gas phase. As we move right, the charges are scaled down, and the distribution shows more of a *gauche* conformation with the *trans* conformation component becoming less dominant. A complete contrast is seen from

the bottom panels, where the RESP-B3LYP parametrization shows predominantly *cis* configurations, which shows very little change as the charges are reduced. In the middle panel, where results from the RESP-HF method are displayed, two things are evident. For full charges, the *cis* conformation dominates while for 0.9 it turns to a *gauche* conformation.

On scaling down further, it reverts to the *cis* conformation. In each of the three charge methods, while the effect of scaling charges affects the electrostatic interactions, all other interactions are unaltered. Hence, we examined in particular the charges on oxygen and fluorine, which could potentially dictate its conformation, and found it to be similar for all charge schemes. If the only influence was due to charges, it would be reasonable to expect a systematic effect in calculated properties.

However, in the liquid environment as the charges are scaled down, the proximity of the ions to one another may increase and various interactions may take place in a manner which is not entirely predictable.

We believe that, just as in case of the *rdfs* discussed earlier, we see here an effect of steric and charge interaction being coupled in RESP-HF/0.9. This might explain the absence of a systematic effect of charge scaling in the RESP-HF method.

To complete this part of the investigation, we also calculated the influence of the charge methods and their scaling on conformation of the cation dihedral C2–N3–C7–C8 and found that distributions are identical regardless of parametrization (Fig. D7 in DiB [89]).

Beside static properties of the pure liquid IL, we further investigate the dynamics by evaluating the self-diffusion coefficient (D) using the Einstein relation:

$$D_i = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \langle [\vec{r}_i(t) - \vec{r}_i(0)]^2 \rangle, \quad (1)$$

where $\langle [\vec{r}_i(t) - \vec{r}_i(0)]^2 \rangle$ is the mean squared displacement (MSD) of the center of mass for species i , $\vec{r}_i(0)$ is the reference position of each particle and $\vec{r}_i(t)$ is its position at time t .

We calculated self-diffusion coefficients using the last 30 ns of production run via ensemble averaging. From Table 1 we see that the cation is more diffusive than the anion, as in most ILs. We also observe a progressive increase in the diffusion coefficients as the charges are scaled down. Since in ILs it is known that the intermolecular interactions between the cation and the anion is stronger than that between two cations or two anions, we can deduce that the effect of charge scaling from 1.0 to 0.9 to 0.85, lowers the strength of the interactions between cation and anion, making them both more mobile. In some sense, the effect is as if the temperature is increased, and is a legitimate approach to mimic dipole interactions in polarizable models. This effect is in tune with the well-known trend that scaling of charges in non-polarizable force fields results in enhanced dynamics for the component ions of the ILs [96]. While, from these results alone, it is difficult to distinguish which charge method performs better, what we do observe is that for all charge methods used in this study, the experimental values lie between those obtained for the scaling factors 0.9 and

0.85. To quantify, both D^+ and D^- calculated for charge methods with no scaling deviate from experimental values by $\sim 70\%$. On the other hand, scaling of charges with S.F. = 0.9 and 0.85 reduce this deviation to approximately 30% and 10% for both cations and anions, respectively.

For the bulk ionic liquid, a careful inspection of static properties showed some marked differences between the results from the three charge methods. CHelpG reproduced expected trends in both *rdfs* and dihedral distribution, while RESP-B3LYP gives a complete contra indication regarding dihedral distribution, with no effect of charge scaling being evident. However, for both *rdfs* and dihedral distribution we see a distinct effect of charge scaling when using RESP-HF with S.F. = 0.9. In the case of ion dynamics in bulk, namely, self-diffusion we see a marked influence of charge scaling for all three charge methods.

V-L-V - surface tension: The interface of the IL on the gas/vacuum boundary plays an extremely important role for its application as a solvent for catalysts, and is a result of the different types of short- and long-ranged interactions related to the complex molecular structure of the IL. Hence, an accurate estimation of the interfacial tension, also known generally as surface tension, is a test field for parametrization of force fields in MD simulations.

Surface tension can be quantified as force per unit length. Given the non-uniformity of local density along the direction normal to the surface, in MD simulations different operational methods and relations for calculation of surface tension have been described in the work of Padua and co-workers [98]. More recently, the so called “Langmuir principle” has been proposed to consider carefully the molecular orientation of ILs at the interface [99].

As demonstrated in Scheme 1 for the V-L-V system, our methodology allows us to properly sample fluctuations of the local density at the surface. Taking advantage of this fact, we calculated the surface tension of the system using the GROMACS tool with the formula:

$$\gamma = -\frac{L_z}{2} \left(\frac{P_{xx} + P_{yy}}{2} - P_{zz} \right), \quad (2)$$

where L_z is length of the box for V-L-V system in the z direction. P_{zz} is the normal pressure P_n and $\frac{P_{xx} + P_{yy}}{2}$ is the tangential pressure $P_t(z)$. Note that Eq. (2) is equivalent to the definition of Irving and Kirkwood. The results are presented for all parametrizations of the force field in Table 2.

We find that the values of surface tension are reasonably reproduced by all force fields and are comparable in magnitude to each other. Of all the nine parametrizations, CHelpG based *ffs* with full charges are the closest to the experimental values [100,101]. On the other hand, RESP-HF and RESP-B3LYP based *ffs* provide slightly lower surface tension values, ranging from 28.9 mN/m to 32.3 mN/m.

The systematic decrease in the values of surface tension with charge scaling is consistent with decreasing the interactions between the ions hence allowing for stronger fluctuations of the interface under the same thermodynamic conditions. A similar effect was observed in experiments when the length of the alkyl

Table 1

Self-diffusion coefficients obtained using MD simulations ($\times 10^{-5}$ cm/s) for different charge methods and for both cation (D^+) and anion (D^-). The experimental value is given in the last row ($\times 10^{-5}$ cm/s) [97].

S.F.	CHelpG		RESP-HF		RESP-B3LYP	
	D^+	D^-	D^+	D^-	D^+	D^-
1.0	0.017	0.011	0.013	0.008	0.018	0.011
0.9	0.049	0.030	0.040	0.024	0.048	0.029
0.85	0.078	0.048	0.065	0.041	0.072	0.042
Exp.	0.059	0.037				

Table 2

Surface tension values (mN/m) obtained using MD simulations for different charge method/S.F. combinations. The experimental value is given in the last row (mN/m) [101].

S.F.	CHelpG	RESP-HF	RESP-B3LYP
	γ	γ	γ
1.0	36.7	32.3	32.0
0.9	32.6	31.4	28.9
0.85	32.0	29.5	29.6
Exp.	35.1		

chain in the cation was increased [101]. It is heartening to know that our values of surface tension for all CHelpG methods, for RESP-HF with scaling factors of 1.0 and 0.9 and for RESP-B3LYP with full charges are still higher than the experimental value for $n = 4$ for $[C_nC_1Im][NTf_2]$ of 30.7 mN/m [101].

S-L-V - X-ray reflectivity: Several techniques, including atomic force microscopy [102,103], infrared spectroscopy [104,105] or X-Ray scattering [106–108], can provide information about SL interfaces buried by a thick layer of a liquid phase, and at the same time yield information on the structural ordering at this interface with sub-molecular accuracy. In our previous work [74] we reported X-Ray reflectivity data at S-L interface which was obtained upon lateral translation of the sample for every data point so as to avoid charging of the surface by prolonged exposure to X-Ray, and averaging over multiple data sets (for details on the experimental procedure see Ref. [74] and the associated SI). The resulting experimental curve is extracted from Ref. [74] and presented in all panels of Fig. 3 as black dots depicting the quantity, R/R_F , which is the measured reflectivity normalized by the Fresnel reflectivity. It can be considered as reliable up to $q = 0.1 \text{ nm}^{-1}$ due to strong noise effects at larger wave numbers. In this reliable range, the curve shows a pronounced minimum at $q = 0.083 \text{ nm}^{-1}$, which is indeed a consequence of interactions with the alumina surface, and is absent in measurements which only probe the IL-vacuum interface [109].

These data were previously used to validate the CHelpG *ff*, along with the Lorentz-Berthelot mixing rule, and at the same time rule out the Kodermann *ff* for modeling IL interfaces [74]. Notably, such comparison of experiments and simulations is challenging due to both uncertainty in the experimental result and the sensitivity of the theoretical modelling. Therefore, the focus is on theoretically reproducing the main feature of the experimental data, namely the minimum at $q = 0.83 \text{ nm}^{-1}$. Furthermore, the previous simulations were performed in the S-L-S configuration. Hence, the modelling has been now improved by the introduction of the vacuum layer to account for a geometry that is closer to the experimental one, and by studying the effect of charge scaling.

To evaluate the computational X-ray reflectivity curves from the prepared S-L-V systems, we follow the previously established procedure [74], and calculate electron densities of cations and anions from the last 30 ns of our MD production runs with a bin size of 0.01 nm. These profiles were convoluted with a Gaussian function of a width $\sigma = 0.16 \text{ nm}$ to simulate the experimental interfacial roughness. The sensitivity of the computational results is shown for the three charge representations (panels Fig. 3a–c), and scaling factor (different lines in each panel).

Considering the left-most panel of Fig. 3, we find that for the CHelpG charge method with S.F. = 1.0, a shallow minimum is obtained at $q = 0.088 \text{ nm}^{-1}$. In the comparison with results in Ref. [74] (see Fig. S2), where the minimum is at $q = 0.093 \text{ nm}^{-1}$, better correspondence with experiments is now obtained. This can be attributed to the introduction of a vacuum interface and presence of an extended bulk region compared to the previous study, neither of which improved the performance of the Kodermann *ff*.

When we further consider the effect of the scaling factor in CHelpG, we see in Fig. 3 (left panel) that, already for S.F. = 0.9, the observed minimum effectively vanishes. Hence, no X-ray reflectivity curve for S.F. = 0.85 is presented. On the other hand, RESP-based charge methods are found to yield X-ray reflectivity curves with profiles matching better to that of the experiment. An unexpectedly deep minimum is observed when RESP-B3LYP is used with no scaling factor (right panel), suggesting overly strong structuring in IL. However, as we scale down from S.F. = 1.0 to 0.9 to 0.85, the reflectivity curve gets progressively flatter. In the case of RESP-HF (middle panel) we observe once again a non-systematic behaviour as charges are scaled. However in all three cases for S.F. = 0.85 the theoretically calculated X-ray reflectivity is flat and devoid of any notable features. From Fig. 3 one can observe that theoretically calculated X-ray reflectivity curves of both RESP-HF and RESP-B3LYP with S.F. = 0.9 match very well to the experimental curve up to $q = 1.1$ which also is the experimentally reliable region [74].

In order to find out how truly representative RESP-HF/0.9 and RESP-B3LYP/0.9 are of the structural ordering of the ionic liquid at the sapphire interface, we calculated electron density data for different lengths of time slices, namely 30, 60 and 100 ns. The effect of such a time averaging allows us to evaluate the stability and the convergence of the solution as a function of the parametrization. From the corresponding curves (see Fig. S3) we find that the basic charge distribution as described within the RESP-HF/0.9 model seems to give a better and a more stable description of the structural correlations in the system than does RESP-B3LYP/0.9.

To summarize, so far in this section we have considered various structural properties, self-diffusion coefficients and surface tension

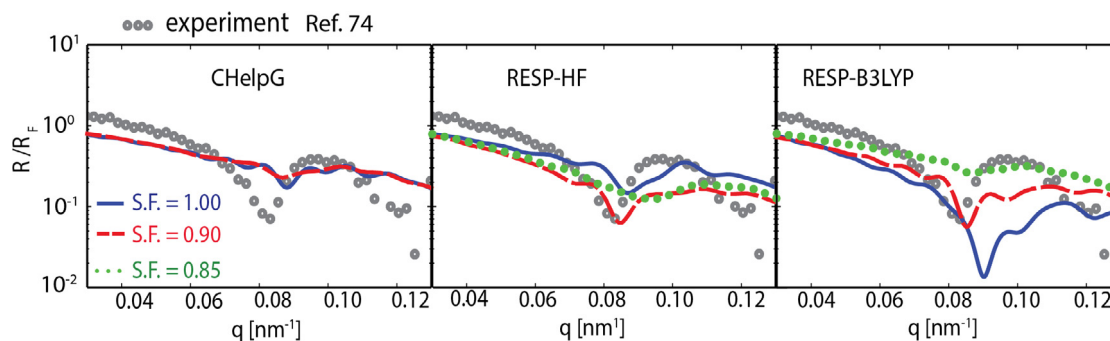


Fig. 3. Comparison of the experimental data extracted from Ref. [74] and theoretically calculated normalized X-ray reflectivity curves as a function of momentum transfer q , for the charge methods (a) CHelpG, (b) RESP-HF and (c) RESP-B3LYP with different scaling factors (S.F. = 1.0 (blue), 0.9 (red) and 0.85 (green)). See text for details. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and examined how well the three charge methods and the three charge scalings describe the IL in its various configurations. While both CHelpG and RESP-HF are well-established and well-tested force fields for ionic liquids, it is interesting to find that RESP-B3LYP, which is well-known for simulating proteins and biological systems, also yielded results of satisfactory, yet somewhat inferior quality. Interestingly enough, for all the three parametrizations a S.F. = 0.85 has been found inadequate in reproducing the experimental properties. On the other hand, when we consider all three charge methods with full charges we find that not only that the mobility of ions is much slower but also the interfacial ordering of the IL with the sapphire substrate is poorer when compared to the experimental results. Since the focus of the study is on understanding the interfacial ordering and dynamics of the IL in the S-L-V system, based on the results of Fig. 3, RESP-HF/0.9 emerges as the natural choice, and will be therefore used henceforth.

3.3. Structural organisation of IL at atomic level

Solid-liquid interface: When considering the structural ordering close to the solid interface, a natural and an interesting question to ask is: at what distance from the substrate do the ions exhibit their bulk liquid properties free from the influence of correlations with the substrate. Note from Scheme 1 that the thickness of the sapphire substrate is 2.12 nm. Here and in all further discussions we define $z = 0$ to start at the top of the surface. A starting point is to generate the INND profile of the S-L-V model system. We find from Fig. 4 that the ordering of both the cation and anion of the IL is pronounced, and persists up to about 6 nm away from the solid surface. Note that in the case of C_2Mim^+ on charged surfaces like mica or silica, or on graphite, fewer layers of structural ordering is observed [110] in contrast to what we observe (cf. Fig. 4). In contrast to the bulk region (Fig. 1), the INND profile close to the S-L interface can be recovered with relatively short sampling times, once equilibration has been achieved, due to strong interactions.

In order to describe the solid-liquid interface, the overall structure of which was described in previous work [74], we now examine both dihedral distributions of the anion and cation conformations as well as the INND per atom type as a function of z at the solid interface. To recall, the $[NTf_2]^-$ with the C–S–S–C dihedral displays *trans/gauche* conformation in the liquid state,

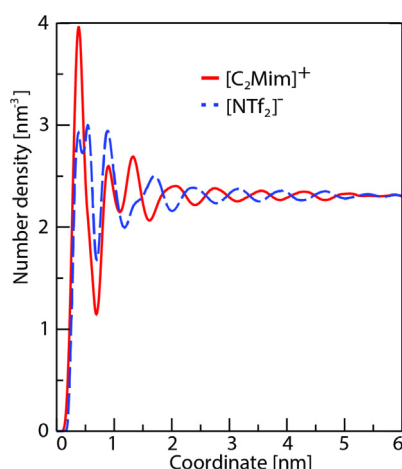


Fig. 4. Interfacial normal number density (INND) profile per molecule of cation (red line) and anion (blue line), close to the S-L interface. The origin of the coordinate system in INND is chosen by the position of hydroxyl groups on the sapphire surface. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

whereas in its crystal phase the *cis* conformation is more dominant [94]. From our computations of the bulk IL, for the case of the RESP-HF/0.9 force field we find that the dihedral distribution of the anions shows a *gauche* conformation (cf. central panel of Fig. 2).

The relevant dihedrals, sampled for 30 ns, are plotted in Fig. 6 and we observe a strong interfacial ordering up to about $z = 0.58$ nm (first maxima in INND). In these first few layers, the dihedral distribution of the anion shows predominately *cis* conformation (top panel, blue line) with all four oxygens pointing towards the surface. However, already between 0.58 and 0.68 nm one observes a conformational transition from *cis* to *gauche* (top panel, orange line), which persists at larger distances. Furthermore, we have also confirmed that the dihedral distribution of the anions continues to have a *gauche* conformation beyond 0.6 nm and into the bulk region. It is indeed surprising that the conformational changes in dihedral distribution occur already for $z > 0.58$ nm, although there is strong ordering of the density of ions until 5 nm (cf. Fig. 4). One clear feature that we observe from snapshots of our MD simulation is the presence of a low density region (minima in INND) between the first few layers and the rest of the liquid at a similar range of z (Fig. D8 in DiB [89] shows a representative snapshot from MD simulations). We believe that this depletion region allows for less restricted internal rotations of the anion, intrinsic to its liquid state above $z > 0.58$ nm. On the other hand, the dihedral of the cation is much less flexible and, unlike for the case of anions, there is no noticeable difference between the distributions calculated in first few layers and those beyond 0.6 nm (see the lower panel in Fig. 5) and this distribution persists into the bulk region of the S-L-V system.

Analogous to the *rdf* plots per atom type (cf. Fig. D5 in DiB [89]), we plot in Fig. D9 in DiB [89] the interface-normal number density per atom type for anion and cations, in order to reveal more precise details of structural organisation of the IL at the solid-liquid interface. From the top panel we see that, in the S-L-V system, oxygens in the anion are closest to the hydroxylated sapphire interface (red line), followed by sulphur (orange line) which is in agreement with the anion dihedral distribution in the *cis* conformation in the first few layers (see top panel Fig. D9 in DiB [89]). Given that C and F atoms are farther from the sapphire interface we can infer that CF_3 groups are pointing towards liquid (cyan and pink).

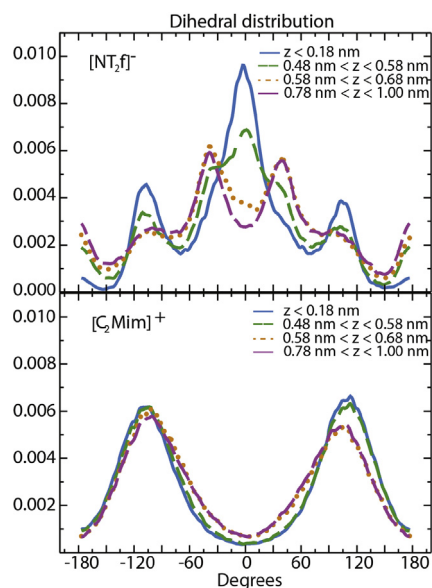


Fig. 5. Dihedral distribution of anions (upper panel) and cations (lower panel) up to 1 nm above the sapphire surface. Note the switch in the conformation of the anion above 0.6 nm.

A corresponding schematic representation of a typical anion conformation is also presented in this panel. In the lower panel we consider, in particular, the orientation of the ring of the cation and find that H9 (see Scheme 1) is closest to the sapphire surface (grey line), followed by the carbon to which it is bonded (light blue line). This is then followed by the nitrogen and carbon atoms of the ring and a corresponding realistic representation of the ring is shown in this panel. A probability distribution analysis of the orientation of the C2–H9 bond, the alignment of the vector connecting the side carbons C6–C7 and the orientation of the ring as a whole, all with respect to the normal of the interface is presented in Fig. D10 in DiB [89]. It turns out that the cation ring is mostly perpendicular to the sapphire surface; the short alkyl chains are mostly parallel to the sapphire interface and hydrogen (H9) points towards the solid surface, all of which confirm the schematic representation of the cation in the lower panel of Fig. D8 in DiB [89].

Further inspection of Fig. D9 in DiB [89] (and Fig. 4) shows that an excess of cations at the surface coexists with anions in the ratio of about 3:4, which induces a 3D bilayer structure that extends deeply toward the bulk. Naturally, even with the excess of positive charge at the sapphire surface, there are almost equal numbers of cations and anions integrated over the first several layers. Despite differences in geometry and force field, an analogous result was also found in our previous work [74] when plotting the 2D-histogram of the centre-of-mass positions of the ions. It was proposed there that since the solid substrate is uncharged, this ordering cannot be understood from a simplistic balance of ionic interactions and that it can be attributed to hydrogen bonding of both cation and anion with the fully hydroxylated sapphire surface [104,105,111].

Liquid-vacuum interface: Before embarking on an investigation of structural organization of the IL on the vacuum interface, it is important to establish that the behaviour of the IL at the vacuum interface is the same, irrespectively of whether we are considering an S-L-V or a V-L-V model system. This consistency is indeed verified in Fig. D11 in DiB [89], where we plot the number density per molecule as a function of z close to the vacuum interface with the sampling time taken to be 160 and 100 ns for S-L-V and V-L-V systems, respectively. Next, we consider how different is the conformation of the anion dihedral in the bulk when compared to that near the liquid-vacuum interface. From the top panel of Fig. 6 we see that a proper bulk region (green) can be defined from around 3 nm away from the vacuum interface (purple). However, from the dihedral distribution displayed in the lower panel of Fig. 6, we see that the conformation at the vacuum interface is very similar to that in the bulk.

Unlike at the S-L interface, there is no real switch in conformation at the L-V interface. Rather, there is only a slightly more pronounced *cis* conformation of the calculated dihedral distribution as we approach the vacuum interface. This agrees well with experimental data at the L-V interface from XPS [112] and LEIS [113] experiments, which show a *cis* conformation followed by *gauche*.

When comparing Fig. 5 with Fig. 6, we see that remarkable changes in the dihedral distribution occur within a nanometre of the S-L interface (see Fig. 5), whereas the changes are much smaller and occur at a larger separation from the L-V interface. In a manner analogous to Fig. 8, we plot the interface-normal number density per atom type for anion and cations in Fig. D12 in DiB [89] to show the structural organisation of the IL at the L-V interface.

Knowing that the vacuum interface offers no real possibility of bonding unlike the solid interface, it comes as no great surprise that in the S-L-V system (top panel of Fig. D12 in DiB [89]), the fluorine atoms of the anion are closest to the vacuum interface (cyan line), while the oxygens (red line) are the farthest from the interface. This implies that oxygens and sulphurs from the anion are mostly buried into ionic liquid while the less polar CF_3 groups

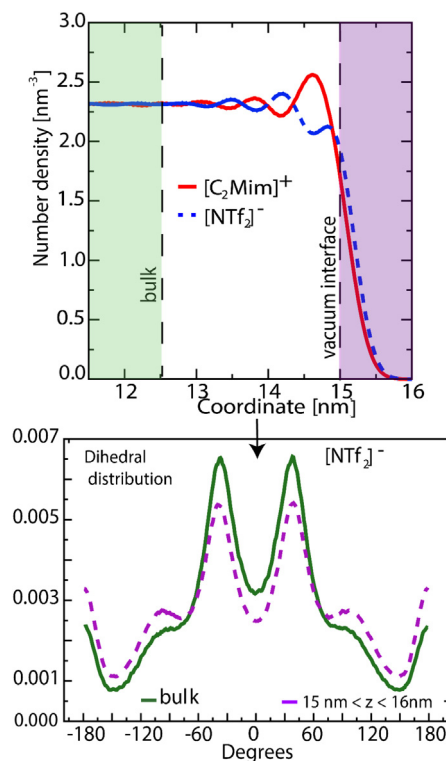


Fig. 6. The upper panel shows the number density profile per molecule as a function of z close to the vacuum interface for the S-L-V system. In the lower panel the p of the dihedral distribution of the anion is displayed for $15 < z < 16$ nm (purple) and for $11.5 < z < 12.5$ nm in the bulk region (green). It is evident that *cis* conformation is more pronounced at the liquid-vacuum interface. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

point towards the vacuum. The slightly lower number density of the carbons (grey line) hints at the presence also of the *gauche* conformation (see schematic representation of the anion). This is in good agreement with the calculated dihedral distribution shown in Fig. 6. From the lower panel of Fig. D12 in DiB [89], we find that it is the alkyl side chain corresponding to C8 (turquoise line) and H17, H18, H19 (magenta line) of the cation that are closest to the vacuum interface, whereas the side chain with C6 (black line) and H12, H13, H14 (blue line) are the farthest removed. This indicates that while the alkyl chain of the cation points into the vacuum side, it is not fully extended above the liquid (see the schematic representation). It has been reported in the literature that as the length of the alkyl side chains increases in imidazolium cations, the alkyl chains which are parallel to the surface normal tend to flip back on to the liquid and show structural ordering, which also results in a decrease of surface tension [114]. However, in our IL as is evident from Fig. D12 in DiB [89], where C_2Mim^+ has the shortest alkyl chain, both cations and anions are in close contact with vacuum interface without any formation of a double layer.

3.4. Dynamics of the IL in S-L-V and V-L-V configuration

The strong structuring effects and the slow dynamics at the S-L interface observed previously [74] points to the fact that careful attention has to be paid to the methodology employed for computing self-diffusion coefficients [115]. Namely, the presence of the solid and/or the vacuum imposes different mobilities parallel (xy) and perpendicular (z) to the interfaces. Furthermore, both components become functions of z . Thus, to evaluate the

mobilities, the V-L-V and S-L-V systems need to be divided into thin slabs of equal thickness along the z -axis. The position of the ions is defined by their centre of mass (COM) coordinates. The optimal thickness of the slabs emerges as a compromise between sampling accuracy and spatial resolution of the diffusion coefficients.

By analysis of the isotropic L system, where the same diffusion coefficient must arise in the slab geometry and from averaging of mean square displacements of all ions in the system, the optimum thickness is found to be 0.5 nm. Besides allowing good statistics from the analysis of 100 ns runs, this also respects the thickness of the interface hydration shells.

Diffusion parallel to the interface as a function of z : As shown in Fig. 7, we computed xy diffusion coefficients using the Einstein relation for 2D (see Eq. (1)) from ensemble-averaged mean square displacement in the x - y plane using the entire 100 ns of production runs so as to have minimal statistical inaccuracies [116]. The horizontal bar in both panels represents the value of the self-diffusion coefficients of the ions of bulk IL (L system) and its spread (cf. Table 1). We find that the solid interface has a dramatic effect on the in plane diffusion coefficient. This is reminiscent the findings of the previous work [74] and from other systems [67,117], whereby the strong interaction with the solid phase freezes the IL surface configuration. In our case, hydrogen bonds between the hydroxylated alumina and both anions and cations are stable on the time scale of tens of nanoseconds. However, even unbound ions in the surface layer experience a significantly reduced diffusion coefficient. Overall, we find diffusion coefficients an order of magnitude smaller than those in the bulk, both for anions and cations (averaging over all ions in the first hydration shell of the alumina). These findings again point to the necessity of long simulations for extracting accurate mobilities at the boundary of the solid and liquid phase.

On the vacuum interface the effects on the mobility are not so dramatic but they are, nonetheless, considerable. Specifically, in-plane diffusion with values of 50% and 100% larger than bulk liquid is obtained for cations and anions, respectively, in the slabs that present a significantly lower concentration than the bulk. It is also worth noting that vacuum interfaces in the V-L-V and the S-L-V systems provide the equivalent results.

Mobility perpendicular to the interface as function of z : For the normal component as a function of z , calculating the mean-square displacement of centres of mass of ions would lead to the tails of probability density distributions being curtailed as the mobility of ions is restricted at the interfaces. Therefore, other methods need to be evoked to extract the mobility [118]. In this

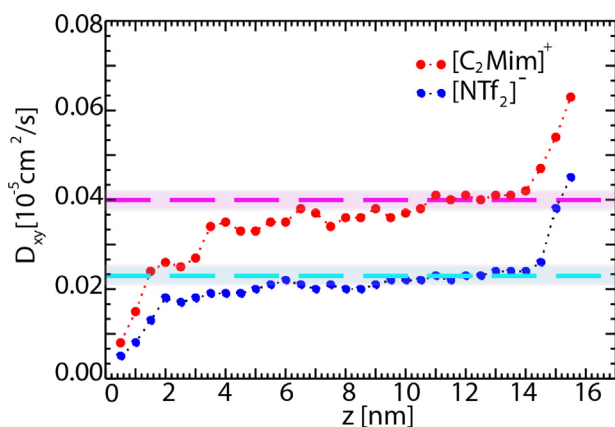


Fig. 7. Lateral diffusion coefficient of cations (red) and anions (blue) in the S-L-V model system as a function of z . Also plotted are bulk values of the diffusion constant for cations (magenta) and anions (cyan) (see Table 1) along with their estimated error bar. See text for details. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

light, we first define and calculate the residence time $t_{res,i}^m(z)$, defined as the time spent by cations and anions in each slab of the simulation cell by simply binning of the position of the COM of the i -th ion over the entire production run and for all z slabs, irrespective of when and how many times the ion visited a particular slab (superscript m denotes the system).

The average over all ions that visited a particular slab $t_{res}^{-m}(z)$ is first calculated for the L system. Given that all z positions in the L system are equivalent, further averaging over all slabs is performed yielding t_{res}^{-L} of 10.328 ± 0.368 ns for the cation, and 13.288 ± 0.28 ns for the anion. Such large values might appear at first sight surprising, and a close examination of trajectories of different ions reveal that even in bulk liquid a given ion does not visit more than 8 or 9 slabs (4–4.5 nm) during the entire 100 ns.

We further proceed with calculating the residence time for the S-L-V system and the V-L-V system, as shown in Fig. 8, where $t_{res}^{VLV}(z)$ and $t_{res}^{-mSLV}(z)$ are presented upon normalisation by t_{res}^{-L} . Interestingly, the residence time reflects both the effect of the crowding and the interactions at the interface. This is evident by the increase of $t_{res}^{VLV}(z)$ and $t_{res}^{-mSLV}(z)$ even in contact with the vacuum, which coincides with the interfacial ordering and the increased density of ions in this particular region of the film (as evident from Fig. D2 in DiB [89]). The very long residence times observed for all three systems are also consistent with very slow equilibrations seen in Fig. 1, where even the vacuum interface equilibrates more slowly than the bulk liquid. This also shows that the residence time is a measure of mixing in the fluid rather than microscopic mobility.

To get an estimate of the mobility of the ions, we therefore calculate the characteristic lifetime $\tau_c^{-m}(z)$ of an ion in the slab (e.g. how long, on average, an ion in a system m , spends in the slab at z before leaving to the neighbouring slab). Practically, using the 100 ns trajectories of each ion, we count the number of times an ion is found in a given slab and record each length of its stay as a separate event. The characteristic time for each slab is found as an average over all events. Notably, $\tau_c^{-m}(z)$ is directly related to the diffusion coefficient, yet, due to the complex structuring of the IL, this relation is not trivial, as evident from the distribution of life times $\tau_c^{-m}(z)$, which, even in the L system, is not an exponential function as would be expected for a simple Brownian particle.

A further difficulty with this approach arises due to inherent noise that the COM experiences due to the internal fluctuations in the ion's internal conformations. A perusal of literature shows

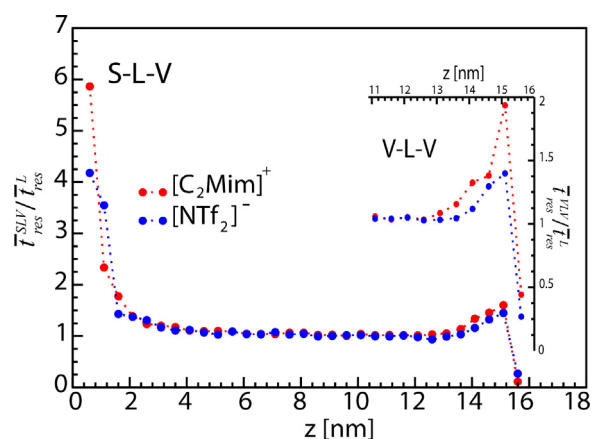


Fig. 8. Normalized residence time of the S-L-V system as compared to the L system as a function of z for cations and anions estimated from 100 ns trajectory in the S-L-V system. In the inset a similar plot for the V-L-V system is shown. Note that the behaviour at the liquid vacuum interface is identical for both S-L-V and V-L-V.

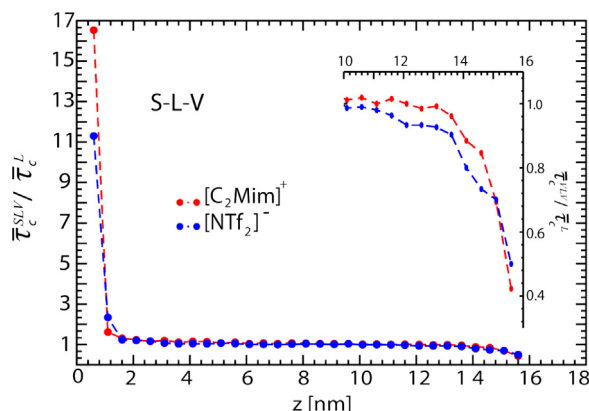


Fig. 9. Normalized characteristic lifetime of S-L-V system as compared to the L system as a function of z for cations and anions estimated from 100 ns trajectory in S-L-V system. In the inset a similar plot for the V-L-V system is shown.

that noise in nanoscale systems with no external fluctuations can be taken to be between 10 and 100 ps [119]. We found, for our system, that a cut-off of 50 ps for the minimum length of the event is essential for obtaining meaningful statistics of true translations close to the solid interface. However, for slabs close to the vacuum interface, a cut-off of even smaller values was found to yield the same estimates for τ_c^{V-L-V} and τ_c^{S-L-V} at appropriate z . This can be understood by the fairly free mobilities of the ions at the L-V interface, while bonding and confinement tend to skew the statistics closer to the S-L interface.

We first calculate τ_c^L (averaged over all slabs) and find values of 0.260 ± 0.003 ns for the cations and 0.301 ± 0.007 ns for the anions, indicating that ions often move between slabs although visiting only a few, as a consequence of stratification in the liquid. Besides, the obtained values of τ_c^L are of the expected order of magnitude for jump diffusion, while the anion to cation ratio is only 1.1, rather than the 1.6 observed in bulk. This suggests that many-body effects significantly affect transport in the IL and particular care has to be given when interpreting mobilities in IL.

Fig. 9 shows $\tau_c^{V-L-V}(z)/\tau_c^L$ and $\tau_c^{S-L-V}(z)/\tau_c^L$. If the inverses of these values are taken as measures of the mobility, one observes that the solid interface slows down the ions by an order of magnitude while the vacuum interface, the ions are more mobile as seen in D_{xy} . Interestingly, the effects on the cations are stronger than on the anions. Furthermore, the comparison of Fig. D2 in DiB [89] and the z -profiles of life times for both ions shows a strong correlation with the effects of structuring. The bulk values are obtained only in the regions where the structure of the fluid in the film with interfaces is indeed that of the bulk liquid. The same effect is clearly visible in $D_{xy}(z)$, $t_{res}^{V-L-V}(z)$ and $t_{res}^{S-L-V}(z)$.

4. Discussion and conclusions

One bottleneck in the development of ionic-liquid based technology is the poor understanding of the fundamental properties of ILs at interfaces, as well as the very limited methodology to study ILs in confined geometries [120]. In our study, we have directly addressed both these issues, firstly by establishing a new methodology, and secondly by applying to a pertinent problem of a widely used IL structuring at the interfaces. We performed an extensive and comprehensive classical molecular dynamics simulation study of the ionic liquid $[C_2Mim][NTf_2]$ in three configurations, namely bulk (L), between two vacuums (V-L-V) and

between a hydroxylated sapphire substrate and a large vacuum (S-L-V) to provide detailed information on structural and dynamic properties of the IL in all three geometries. However, to achieve this goal we addressed a major challenge to find an optimal force field within the framework of non-polarizable force fields, which can address all three geometries with the same level of accuracy. This is important because force fields are typically benchmarked for properties of bulk liquids [52–65]. Furthermore, although the thrust in this field might be moving towards use of polarizable force-fields, for a wide community of researchers, finding optimal non-polarizable force-fields is of great importance, especially in light of the associated computational costs. Therefore, we first focused on finding a force-field parametrization that equally accurately models bulk liquids and liquids in confined geometries of the interfaces.

Using our prior-established knowledge [74], we studied the suitability of three force-fields, using three charge scaling methods for each. We performed an optimization of force field parameters simultaneously, for all three configurations, against all experiments available in the literature for the ionic liquid of choice. Indeed, we found that for our chosen IL, two of the three typical force fields that are acceptable in bulk do not always work at interfaces, and RESP-HF/0.9 emerged as the optimal force-field. This is primarily based on its ability to reproduce experimental results, including the surface tension, diffusivity and structure factors, and especially those relevant to the S-L-V system, such as X-ray reflectivity at the S-L interface [74], and molecular orientations measured by XPS [112] and LEIS [113] at the L-V and S-L-V interface. Thus, we believe this is the first comprehensive effort in prototypical simulation set-up applicable to all geometries for simulating complex liquids, and is a marked improvement in the field [67].

Other key factors emerged as part of this exercise. In order to realistically model the system so as to preserve the bulk behaviour at regions away from the interfaces, we quantified the optimally required large system sizes and long simulation times needed for each of the three geometries. We showed that 4–6 nm from the interfaces, the confined liquids become structurally and even dynamically equivalent to infinite bulk systems, although proper mixing and the convergence of thermodynamic properties takes twice as long in bulk of a V-L-V film, and three times longer in the bulk of the S-L-V geometry, where runs of 250 ns were necessary for the given systems size. The 200 ns long sampling of reasonably large V-L-V systems also permits a direct accurate determination of the surface tension, without introducing additional constraints [98,99]. Finally, to recover the correct density stratification represented by INND profiles at both interfaces of S-L-V geometry, large systems are necessary so that a sufficient bulk region exists to prevent any influence of the solid/vacuum interface on the ordering of ions at vacuum/solid interface. Furthermore, the imposition of a large vacuum for the S-L-V system helps to remove artefacts of electrostatic interactions along the z -direction [87,88], so that the influence of charge scaling on the interfacial ordering of the IL at the substrate could be examined with confidence.

Apart from above results that are intrinsically interesting and help to validate the methods employed, the following physical insights have been obtained. To start with, detailed information at molecular level of structural organisation at the interfaces could be gained. Specifically, we could show that the internal conformational distribution of the anions is spatially dependent and can be strongly influenced by the interface itself. In contrast to bilayer ordering of ions of IL found at charged surfaces [106], here the complex 3D layering of cations and anions of the IL on the hydroxylated sapphire substrate is found to be a checker-board arrangement, due to the role played by hydrogen bonding.

Our structural investigation was complemented by a comprehensive study of the diffusion of the ions in *xy* and *z* directions in the S-L-V system that features both interfaces. Given the inhomogeneous environment of the interfaces, whose effects are distinct on the overall mixing of the confined liquid as well as on molecular translations, we find that both in their lateral diffusion and along *z*, the ions become very slow at the S-L interface, but are only mildly affected by the presence of the vacuum. A strong correlation between the stratification and dynamics in the interfacial layers is detectable deep into the films, pointing to the importance of many-body interactions in defining transport properties in the IL.

Furthermore, the choice of IL, [C₂Mim][NTf₂], was dictated by it being a most commonly used IL in technological applications today. Where there was until now no general model to deal with its behaviour in multiphase systems, a solid basis has been provided by our work. This major fundamental advance in the field is a necessary requirement for future studies of reactive IL films, examples of which are SILP [34] or SCILL catalysis [37], separation techniques or sequencing technologies based on solid-state pores [121]. Consequently, our results can be directly transferred to the realm of other applications in this IL.

We note that this detailed study has validated our hypothesis that a prototypical approach using MD simulations can lead to a full understanding of the static and dynamic properties of the IL in bulk and confined environment. This protocol of system setup, system size and simulation times, as well as search for optimal force-field parameters can also be exploited in studies on other imidazolium based ILs and other substrates for various applications of ILs embedded in films [26] or in gelators [21].

Using this body of work as a solid framework for further studies on our model systems, we plan to examine the structural organization in the entire family of cations with increasing length of alkyl chains in an S-L-V model system and examine dynamics using a jump-diffusion type of models. A more exciting line of study which is of the importance to interfacial science both from fundamental as well as application point of view is to understand the structural ordering of mono- and bi-layers of the IL depending on how they are prepared from a drop in wetting simulations. In view of the fact that the basic mechanism for the Water-Gas Shift Reaction of Ruthenium-based SILP catalysts was recently established using quantum calculations [122], we plan to elaborate on this mechanism using MD simulations by including an atomistic description of the IL and to study the dynamics of the gases and this complex incorporated in our S-L-V model system.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2019.06.017>.

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